Hao Li

Postdoctoral Researcher

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Homepage

Profile

My research focuses on design engineering based on mathematics, computational mechanics, and informatics, with expertise in numerical methods for multiphysics simulation, method development in topology optimisation, and high-performance computing. I am particularly interested in the optimum design of multi-scale and multiphysics systems, such as heat exchangers and microchannel cooling devices, and their applications in industrial products. At present, I am extending my research towards network scheduling optimisation, such as district heating systems, coupled with integrated assessment modelling. Alongside research, I lecture on finite element analysis and structural optimisation and supervise Ph.D. and M.Sc. projects.

Current Position

2025/10–now **Postdoctoral Researcher**, MSOE Group, Inst. Mech. Electr. Eng., SDU, Denmark.

Professional Experience

Research Positions

2023/10–25/09 **MSCA Postdoc Researcher**, Inst. Mech. Electr. Eng., SDU, Denmark — Supervisor: Joe Alexandersen
2023/04–23/09 **Specific Researcher**, Dept. Micro Eng., Kyoto Univ., Japan — Supervisor: Shinji Nishiwaki
2022/10–23/03 **JSPS Postdoc Researcher**, Dept. Mech. Eng. Sci., Kyoto Univ., Japan — Supervisor: Kazuhiro Izui

Research Stays

2026/03 **Visiting Scholar**, Dept. Chem. Eng., TU/e, Netherlands — Host: Antoni Forner-Cuenca & Mojtaba Barzegari
2025/06–25/07 **Visiting Scholar**, LIP 6, Sorbonne Univ., France — Host: Pierre Jolivet
2024/10–24/12 **Visiting Scholar**, Div. Appl. Math., Brown Univ., USA — Host: George EM Karniadakis
2018/07–18/10 **Research Assistant**, Dept. Dyn. Mech., Universität Paderborn, Germany — Supervisor: Paul Dunst

Teaching Activities

Mean score of student evaluations is indicated in parentheses (if any).

2024–now **Lecturer**: Inst. Mech. Electr. Eng., SDU.
- Engineering Optimisation (undergraduate)
- Structural Optimisation: Theory and Implementation (graduate) (3.67/4.0)
- Advanced Finite Element Analysis (graduate)
- Finite Element Analysis (undergraduate) (3.75/4.0)

2022–2023 **Guest Lecturer**: Dept. Mech. Eng. Sci., Kyoto Univ.
- Advanced Finite Element Method (graduate)

Education

2022/09 **Ph.D.**, *Mechanical Engineering and Science,* Kyoto University, Japan.
- *Advisor*: Prof. Shinji Nishiwaki
- *Co-advisor*: Prof. Kazuhiro Izui
- *Dissertation*: Level set-based topology optimization of thermal fluid–structure systems. 10.14989/doctor.k24226.

2019/06 **M.Sc.**, *Mechanical Design,* University of Shanghai for Science and Technology, China.
- *Advisor*: Prof. Xiaohong Ding
- *Co-advisor*: Prof. Dalei Jing
- *Dissertation*: High heat flux cooling device designed by topology optimization. 10.27308/d.cnki.gslgu.2020.000010.

2016/06 **B.E.**, *Mechanical Design, Manufacture & Automation,* University of Shanghai for Science and Technology, China, graduate student admitted through recommendation

2016/06 **B.E.** (First-Class Honours), *Manufacturing Systems Engineering,* Liverpool John Moores University, UK.
- *Advisor*: Dr. Adrian Chong (senior lecturer)
- *Dissertation*: To build an automated system to straighten the annealed swayed copper pipe of the electronic expansion valve.

Funding

Principal Investigator

2023/10–25/09 **Marie Skłodowska-Curie Actions (MSCA) Postdoc Fellowship**, Grant-in-aid, Project title: advanced design of heat exchangers using multiscale models and machine learning (ADeHEx), €214,934, principal investigator. 10.3030/101106842. [News]

2023/04–23/09 **Denso Corporation Heat Exchanger R&D Dept. Industrial Project**, Grant-in-aid, Project title: topology optimization of heat exchanger and development of its user interface, ¥2,146,000 (€133,900), principal investigator. [Link]

2021/04–23/03 **Japan Society for the Promotion of Science (JSPS) Doctoral Fellowship**, Grant-in-aid, Project title: large-scale topology optimization of thermal fluid-structure system design, ¥5,760,000 (€34,800) as salary, ¥1,500,000 (€11,700) as research fund, principal investigator. Grant ID: 21J13418

2018/07–18/09 **Deutscher Akademischer Austauschdienst (DAAD)**, Grant-in-aid, Project title: Experimental study on the dynamic behaviour of piezoelectric ultrasonic transducer, €1,950, principal investigator.

Contributor

2026/05–26/10 **CORINTIS Industrial Project**, Grant-in-aid, Project title: multi-scale topology optimisation of microchannel cooling devices, principal investigator: Joe Alexandersen. My role: main contributor. [Link]

2024/10–25/12 **Centre Technique Industriel de la Plasturgie et des Composites (CTIPC) Industrial Project**, Grant-in-aid, Project title: topology optimisation of conformal cooling channels in injection moulding, principal investigator: Yupeng Sun & Joe Alexandersen. My role: code package developer (joined in 2025/10). [Link]

2023/10–27/09 **Independent Research Fund Denmark's Sapere Aude grant**, Grant-in-aid, Project title: Breaking the time barrier: COmputational Morphogenesis FOR Time-dependent problems (COMFORT), Total grant: 6,000,000 DKK (€805,000), principal investigator: Joe Alexandersen. My role: contributor (joined in 2025/10). [News]

2018/01–19/12 **Shanghai Aerospace Science and Technology Innovation Fund Projects**, Grant-in-aid, Project title: Structural bionic design for cold plate used in the high heat flux electronic device, ¥100,000 (€14,700), principal investigator: Xiaohong Ding. My role: main contributor.

Travel Grants

2024 **Otto Mønsteds Fond**, Grant-in-aid, research visits, 30,000 DKK (€4,013). [Link]

Prizes & Awards

[4] **ISSMO/SPRINGER Prize 2025**, International Society for Structural and Multidisciplinary Optimization (ISSMO), 2025. Awarded for the paper titled "Homogenisation-based topology optimisation for convection-dominated heat transfer incorporated with neural network surrogate modelling." [News]

[3] **Professor Avram Bar-Cohen Best Paper Award**, 24th Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm), IEEE, 2025. Awarded for the paper titled "Phasor-based Dehomogenisation for Microchannel Cooling Topology Optimisation." [News]

[2] **Outstanding Master Thesis Award**, University of Shanghai for Science and Technology, 2019. Awarded for the dissertation titled "High heat flux cooling device designed by topology optimization." [Link]

[1] **Outstanding Paper Award**, 18th International Conference of Electronic Packaging Technology & High Density Packaging (ICEPT), IEEE, 2017. Awarded for the paper titled "Design of microchannel heat sink using topology optimization for high power modules cooling."

Granted Patents

[1] Ding, X., Ji, Y. and **Li, H.** (2017). "Bionic heat flow channel design method". China Patent ZL 2017 1 0373075.8. China National Intellectual Property Administration.

Academic Services

Peer Review Activities

(Number of manuscripts reviewed is indicated in parentheses.)

- Int. J. Heat Mass Transf. (8)
- Comput. Model. Eng. Sci. (1)
- Struct. Multidiscip. Optim. (22)
- Electron. Res. Arch. (1)
- Int. J. Numer. Methods Fluids (1)
- Nucl. Eng. Des. (1)
- Sci. Rep. (1)
- Int. Commun. Heat Mass Transf. (4)
- Addit. Manuf. (1)
- Appl. Math. Model. (1)
- Compos. Struct. (1)
- Eng. Optim. (1)
- Comput. Methods Appl. Mech. Eng. (11)
- Comput. Math. Appl. (1)
- J. Mech. Des. (1)
- J. Energy Storage (2)
- Eng. Struct. (1)
- Eng. Comput. (1)

- Adv. Eng. Softw. (2)
- J. Appl. Mech. (1)
- Appl. Therm. Eng. (2)
- Int. J. Heat Fluid Flow (1)
- Comput. Struct. (2)
- JOSS (1)
- IEEE ITherm (3)
- Finite Elem. Anal. Des. (1)

Memberships of Academic Societies

2023–now The Danish Center for Applied Mathematics and Mechanics (DCAMM)
2022–2023 The Japan Society of Mechanical Engineers (JSME)
2020–2023 The Japan Society for Industrial and Applied Mathematics (JSIAM)
2020–now Asian Society for Structural and Multidisciplinary Optimization (ASSMO)
2019–now International Society for Structural and Multidisciplinary Optimization (ISSMO)

Conference & Minisymposium Organisation

[1] Minisymposium "Topology, Shape and Parameter Optimisation for Time-Dependent Systems", Track: Inverse Problems, Optimization and Design, WCCM-APCOM-ECCOMAS 2026, 17th World Congress on Computational Mechanics and 10th European Congress on Computational Methods in Applied Sciences and Engineering, Munich, Germany, July 19–24, 2026. Main organiser: Joe Alexandersen, my role: co-organiser.

Scientific & Pedagogical Skills

Scientific Skills

Topology optimisation of multi-scale and multiphysics systems. I apply and extend a broad range of topology optimisation methods, including level-set and density-based approaches, as well as homogenisation-based multi-scale design with dehomogenisation, to the optimum design of thermal-fluid devices.

Numerical methods for multiphysics simulation. I formulate and solve partial differential equations (PDEs) governing fluid–structure interaction, conjugate heat transfer, and space-time heat conduction, using distributed finite element solvers with domain decomposition, preconditioners, and adaptive mesh refinement (PETSc/SLEPc, MMG/ParMMG). I maintain easy-to-use open-source frameworks in FreeFEM and Firedrake, enabling high-resolution 3D design in both academic and industrial applications.

Scientific machine learning. I build neural-network surrogate models in PyTorch and integrate them into physics-based design workflows, such as homogenisation-based topology optimisation for convection-dominated heat transfer. I also have hands-on experience with Physics-Informed Neural Networks (PINNs) and Deep Operator Neural Networks (DeepONet), gained during my research stay at the CRUNCH group, Brown University.

Experimental design and validation. I design and conduct experiments on liquid-cooled channels to validate numerical simulations and bridge the gap between simulation and real-world prototypes.

Pedagogical Skills

Teaching. I lecture on four courses in Mechanical Engineering spanning undergraduate and graduate levels, and I am involved in designing the course content; my courses receive consistently high student evaluations. My teaching philosophy focuses on creating an interactive learning environment where students engage with complex concepts through hands-on exercises, MATLAB code demonstrations, and projects. I aim to foster curiosity and critical thinking by combining theory with practical application, refining my teaching continuously through student feedback.

Supervision. I co-supervise one Ph.D. project and have completed the co-supervision of two M.Sc. thesis projects on topology optimisation of cooling devices, and coach student teams for the IEEE ITherm Cold Plate Student Design Competition (2025, 2026). I am open to exploring new and emerging topics together with students, e.g., a Master's course project on LLM-assisted FEA, an idea developed jointly with the students. I foster a collaborative environment where students take initiative in their work, balancing technical guidance with encouraging independent thinking to help them complete high-quality research.

Student Supervision

Ph.D. & M.Sc. Projects

2025–2026 Jacob Mads Sakari Andersen, Christian Hertel Larsen, SDU. M.Sc. project: "Conjugate heat transfer topology optimization for cooling devices on Formula Student race car using Firedrake."
2024–2026 Amirhossein Bayat, SDU. Ph.D. project: "Topology optimisation for high heat flux cooling."
2024–2025 Casper Skjold Buhrkall, SDU. M.Sc. project: "Development of topology optimisation framework for design of liquid-cooled heat sinks."

Semester Projects & Competitions

2026 Rasmus Ottar Kobayashi Olsen, Lukas Petersen, SDU. Theory and method for research and development work
2026 Amirhossein Bayat, Magnus Appel, SDU. IEEE ITherm 2026 – Cold Plate Student Design Competition
2025 Amirhossein Bayat, Magnus Appel, SDU. IEEE ITherm 2025 – Cold Plate Student Design Competition

International Collaborations

(Listed alphabetically by family name.)

Mojtaba Barzegari, Postdoc Researcher at TU/e: Computational and mathematical modelling of biodegradation and electrochemical systems. Host of my research stay at TU/e (2026).

Li Chen, Professor at Xi'an Jiaotong University: Recent advances in thermal-fluidic topology optimisation. I serve as a consultant for the 6th edition of the "Heat Transfer" textbook.

Yunfei Fu, Professor at Shandong University of Science and Technology: Level set-based topology optimisation. I contributed numerical examples to his book "Smooth Topological Design of Continuum Structures" (CRC Press, 2025), initiated during his time at University of Alberta.

Simon Garnotel, Multiphysics simulation engineer at Airthium: Industrial collaboration on the TANATLOC GUI for FreeFEM-based conjugate heat transfer optimisation, resulting in 1 joint publication.

Pierre Jolivet, Professor at LIP 6, Sorbonne Université: Distributed finite element analysis and high-performance computing. Co-author of 7 joint publications; host of my research stay at Sorbonne Université (2025).

George EM Karniadakis, Professor at Brown University: Physics-Informed Neural Networks (PINNs). Host of my MSCA secondment research stay at the CRUNCH group (2024).

Tsuguo Kondoh, Research Scientist at Kyoto University: Topology optimisation of fluid-thermal problems. Long-term collaborator since my Ph.D.; co-author of 6 joint publications.

Hui Liu, Associate Professor at Wuhan University: Parametrised topology optimisation for fluidic systems. Co-supervision of a Bachelor thesis, resulting in 1 joint publication.

Hiroshi Ogawa, Project Manager at Denso Corporation: Topology optimisation of heat exchangers. Industrial collaboration on the manufacturing and experimental validation of optimised prototypes, resulting in 1 joint publication.

Masaki Otomori, Project Manager at AISIN Corporation: Topology optimisation of insulating shields in the electroplating process using tertiary current distribution. Continuous collaboration since my departure from Kyoto University, resulting in 1 joint publication.

Renato Picelli, Associate Professor at University of São Paulo: Body-fitted topology optimisation of binary structures. Collaboration with his Ph.D. candidate, resulting in 1 joint publication.

Atsushi Suzuki, Guest Associate Professor at Cybermedia Center, Osaka University: Topology optimisation of heat exchangers. External technical support on numerical solvers for our research group.

Zhicheng Wang, Associate Professor at Brown University: PINNs for multi-scale conjugate heat transfer problems. Collaboration within the CRUNCH group during my research stay.

Takayuki Yamada, Associate Professor at the University of Tokyo: Level set-based topology optimisation methods. Co-author of 3 joint publications.

Heng Zhang, Associate Professor at University of Shanghai for Science and Technology: Multiscale topology optimisation of dynamic problems and biomechanical designs. Long-term collaborator since my M.Sc.; co-author of more than 10 joint publications.

Bengliang Zhu, Professor at South China University of Technology: Topology optimisation of compliant mechanisms and feature-driven methodologies. Co-author of 7 joint publications.

Publication Overview

Google Scholar: Publications: 50 - Citations: 1,454 (mean: 29.0) - h-index: 19 - i10-index: 31 [Link]

Web of Science: GNP-0695-2022 Publications: 40 - Citations: 1,088 (mean: 27.2) - h-index: 16

Scopus: 57201868811 Publications: 43 - Citations: 1,218 (mean: 28.3) - h-index: 17

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First/corresponding author: 11 peer-reviewed journal articles (average citations: 74.5), 3 conference/technical papers, 23 conference/webinar presentations

(as of July 14, 2026)

Peer-Reviewed Journal Publications

Under Review & Submitted

[1] **Li, H.** and Alexandersen, J. (2026). "Homogenisation-based topology optimisation for convection-dominated heat transfer with flow-dependent effective properties." Preprint submitted to *Optimization and Engineering*. hal.science/hal-05225454v2

Published & Accepted

[43] Bayat, A., **Li, H.**, and Alexandersen, J. (2026). "Density-based topology optimization for turbulent fluid flow using the standard k-ε RANS model with wall-functions imposed through an implicit wall penalty formulation." *Computer Methods in Applied Mechanics and Engineering*, 460, 119100. 10.1016/j.cma.2026.119100.

[42] Otomori, M., **Li, H.**, Ishida, N., Izui, K., and Nishiwaki, S. (2026). "Topology optimization for the design of insulating shield on electroplating process using tertiary current distribution." *Structural and Multidisciplinary Optimization*, 69, 101. 10.1007/s00158-026-04301-1.

[41] Bayat, A., **Li, H.**, and Alexandersen, J. (2026). "Density-based topology optimization for Navier-Stokes flow with free-slip boundary conditions." *Structural and Multidisciplinary Optimization*, 69, 23. 10.1007/s00158-025-04212-7.

[40] Hu, T., Wang, Y., **Li, H.**, Noguchi, S., Zhang, H., Furuta, K., Izui, K., and Nishiwaki, S. (2025). "Concurrent multiscale topology optimization of coated structures infilled with multiple self-connected microstructures." *International Journal for Numerical Methods in Engineering*, 126(21). 10.1002/nme.70173.

[39] **Li, H.**, Garnotel, S., Jolivet, P., Ogawa, H., Kondoh, T., Furuta, K., Alexandersen, J. and Nishiwaki, S. (2025). "3D topology optimization of conjugate heat transfer considering a mean compliance constraint: advancing toward graphical user interface and prototyping." *Advances in Engineering Software*, 207, 103939. 10.1016/j.advengsoft.2025.103939.

[38] **Li, H.**, Jolivet, P. and Alexandersen, J. (2025). "Multi-scale topology optimisation of microchannel cooling using a homogenisation-based method." *Structural and Multidisciplinary Optimization*, 68(8). 10.1007/s00158-024-03931-7.

[37] Sasaki, T., Furuta, K., Ishida, N., **Li, H.**, Kondoh, T., Izui, K. and Nishiwaki, S. (2024). "Topology optimization for 3D fluid diode system design considering self-supporting property." *Structural and Multidisciplinary Optimization*, 67, 209. 10.1007/s00158-024-03920-w.

[36] Lu, Y., Chang, X., Zhang, Z., Zhou, Y., Liu, H. and **Li, H.** (2024). "A body-fitted adaptive mesh and Helmholtz-type filter based parameterized level-set method for structural topology optimization." *Acta Mechanica Sinica*, 41, 424119. 10.1007/s10409-024-24119-x.

[35] Hu, T., Wang, Y., **Li, H.**, Yu, M., Furuta, K., Izui, K. and Nishiwaki, S. (2024). "Topology optimization of coated structures infilled with multiple materials." *Finite Elements in Analysis & Design*, 235, 104165. 10.1016/j.finel.2024.104165.

[34] Azevedo, A.S.C., **Li, H.**, Ishida, N., Siqueria, L., Cortez, R.L., Silva, E.C.N., Nishiwaki, S. and Picelli, R. (2024). "Body-fitted topology optimization via integer linear programming using surface capturing techniques." *International Journal for Numerical Methods in Engineering*, 125(13), e7480. 10.1002/nme.7480.

[33] Fujimoto, Y., Furuta, K., Kondoh, T., **Li, H.**, Izui, K. and Nishiwaki, S. (2024). "Optimum design method for structural configuration and fiber arrangement for fiber-reinforced composites." *Composite Structures: Part C*, 13, 100432. 10.1016/j.jcomc.2024.100432.

[32] Xiong, M., Yang, J., Ding, X. and **Li, H.** (2023). "Topology optimization design of micromixer based on principle of Tesla valve: an experimental and numerical study." *Chemical Engineering and Processing - Process Intensification*, 193, 109560. 10.1016/j.cep.2023.109560.

[31] Hu, T., Wang, Y., Zhang, H., **Li, H.**, Ishida, N., Furuta, K., Izui, K. and Nishiwaki, S. (2023). "Concurrent multiscale topological design of coated structures with layer-wise bi-material lattice infill for minimum dynamic compliance." *Composite Structures*, 323, 117468. 10.1016/j.compstruct.2023.117468.

[30] **Li, H.**, Yu, M., Jolivet, P., Alexandersen, J., Kondoh, T., Hu, T., Furuta, K., Izui, K. and Nishiwaki, S. (2023). "Reaction-diffusion equation driven topology optimization of high-resolution and feature-rich structures using unstructured meshes." *Advances in Engineering Software*, 108, 103457. 10.1016/j.advengsoft.2023.103457.

[29] Nakayama, N., **Li, H.**, Furuta, K., Nishiwaki, S. and Izui, K. (2023). "Maximization of eigenfrequencies using topology optimization based on multi-material level set method." *Mechanical Engineering Journal*. 10.1299/mej.22-00353.

[28] Zhang, H., Ding, X., Guo, H., Xu, S., **Li, H.**, Nishiwaki, S., Chen, Y. and Takezawa, A. (2023). "Multiscale topology optimization of biodegradable metal matrix composite structures for additive manufacturing." *Applied Mathematical Modelling*, 114, 799–822. 10.1016/j.apm.2022.10.019.

[27] Yeranee, K., Rao, Y., Li, Y. and **Li, H.** (2022). "Improved thermal performance of a serpentine cooling channel by topology optimization infilled with triply periodic minimal surfaces." *Energies*, 15(23). 10.3390/en15238924.

[26] **Li, H.**, Kondoh, T., Jolivet, P., Nakayama, N., Furuta, K., Zhang, H., Zhu, B., Izui, K. and Nishiwaki, S. (2022). "Topology optimization for lift–drag problems incorporated with distributed unstructured mesh adaptation." *Structural and Multidisciplinary Optimization*, 65, 222. 10.1007/s00158-022-03314-w.

[25] Ishida, N., Kondoh, T., Furuta, K., **Li, H.**, Izui, K. and Nishiwaki, S. (2022). "Topology optimization for maximizing linear buckling load based on level set method." *Mechanical Engineering Journal*, 9(4), 21-00425. 10.1299/mej.21-00425.

[24] Hu, T., Wang, Y., Zhang, H., **Li, H.**, Ding, X., Izui, K. and Nishiwaki, S. (2022). "Topology optimization of coated structures with layer-wise graded lattice infill for maximizing the fundamental eigenfrequency." *Computers & Structures*, 271, 106861. 10.1016/j.compstruc.2022.106861.

[23] Yeranee, K., Rao, Y., Li, Y. and **Li, H.** (2022). "Enhanced thermal performance of a pin-fin cooling channel for gas turbine blade by density-based topology optimization." *International Journal of Thermal Sciences*, 181, 107783. 10.1016/j.ijthermalsci.2022.107783.

[22] Pan, S., Yu, M., **Li, H.**, Li, Z., Reng, M., Gu, J. and Shen, C. (2022). "An integrated two–step strategy for an optimal design of liquid–cooled channel layout based on the MMC-Density approach." *Structural and Multidisciplinary Optimization*, 65, 221. 10.1007/s00158-022-03315-9.

[21] Zhang, H., Takezawa, A., Ding, D., Zhang, X., Xu, S., **Li, H.**, Nozawa, S. and Nishiwaki, S. (2022). "Robust topology optimization of biodegradable composite structures under uncertain degradation rates." *Composite Structures*, 291, 115593. 10.1016/j.compstruct.2022.115593.

[20] Meng, F., Ding, X., **Li, H.** and Xiong, M. (2022). "Optimal design of liquid-cooled uniform-temperature plate with hierarchical vein structure." *Journal of Mechanical Engineering*, 58(22), 426–437. (in Chinese). 10.3901/JME.2022.22.426.

[19] **Li, H.**, Kondoh, T., Jolivet, P., Furuta, K., Yamada, T., Zhu, B., Zhang, H., Izui, K. and Nishiwaki, S. (2022). "Optimum design and thermal modeling for 2D and 3D natural convection problems incorporating level set-based topology optimization with body-fitted mesh." *International Journal for Numerical Methods in Engineering*, 123(9), 1954–1990. 10.1002/nme.6923.

[18] Zhu, B., Wang, R., Liang, J., Lai, J., Zhang, H., Li, H., **Li, H.**, Nishiwaki, S. and Zhang, X. (2022). "Design of compliance mechanism: an explicit topology optimization method using end-constrained spline curves with variable width." *Mechanism and Machine Theory*, 171, 104713. 10.1016/j.mechmachtheory.2021.104713.

[17] Zhang, H., Ding, X., Chen, Y., Zhang, X., **Li, H.** and Ni, W. (2022). "Concurrent topology optimization of composite plates for minimum dynamic compliance." *Materials*, 15(2), 538. 10.3390/ma15020538.

[16] Hu, T., Ding, X., Zhang, H., Shen, L. and **Li, H.** (2023). "Geometry and size optimization of stiffener layout for three-dimensional box structures with maximization of natural frequencies." *Chinese Journal of Aeronautics*, 36(1), 324–341. 10.1016/j.cja.2022.06.010.

[15] **Li, H.**, Kondoh, T., Jolivet, P., Furuta, K., Yamada, T., Zhu, B., Izui, K. and Nishiwaki, S. (2022). "Three-dimensional topology optimization of fluid-structure system using body-fitted mesh adaption based on the level-set method." *Applied Mathematical Modelling*, 101, 276–308. 10.1016/j.apm.2021.08.021.

[14] Zhang, H., **Li, H.**, Ding, X., Hu, T., Pan, S., Ni, W. and Xu, S. (2022). "Numerical study on the body-fitted topology optimization for three-dimensional high resolution structure design." *Journal of Mechanical Engineering*, 58(5), 136–143. (in Chinese). 10.3901/JME.2022.05.136.

[13] Zhu, B., Wang, R., Zhang, H., Li, H., Liang, J., Zhang, X., **Li, H.** and Nishiwaki, S. (2022). "An approach for geometrically nonlinear topology optimization using moving wide-Bézier components with constrained ends." *Journal of Mechanical Design*, 144(1), 011704. 10.1115/1.4051872.

[12] Zhang, H., Takezawa, A., Ding, X., Xu, S., Duan, P., **Li, H.** and Guo, H. (2021). "Bi-material microstructural design of biodegradable composites using topology optimization." *Materials & Design*, 209, 109973. 10.1016/j.matdes.2021.109973.

[11] Zhang, H., Takezawa, A., Ding, X., Xu, S., **Li, H.** and Guo, H. (2021). "Topology optimization of degradable composite structures with time-changeable stiffness." *International Journal for Numerical Methods in Engineering*, 122, 4751–4773. 10.1002/nme.6745.

[10] Zhu, B., Zhang, X., Li, H., Wang, R., Liu, M. and **Li, H.** (2021). "Topology optimization of multi-material compliant mechanisms using node-density interpolation scheme." *Journal of Mechanical Engineering*, 57(15), 53–61. (in Chinese). 10.3901/JME.2021.15.053.

[9] **Li, H.**, Yamada, T., Jolivet, P., Furuta, K., Kondoh, T., Izui, K. and Nishiwaki, S. (2021). "Full-scale 3D structural topology optimization using adaptive mesh refinement based on level-set method." *Finite Elements in Analysis and Design*, 194, 103561. 10.1016/j.finel.2021.103561.

[8] Zhu, B., Wang, R., Wang, N., **Li, H.**, Zhang, X. and Nishiwaki, S. (2021). "Explicit structural topology optimization using moving wide Bézier components with constrained ends." *Structural and Multidisciplinary Optimization*, 64, 53–70. 10.1007/s00158-021-02853-y.

[7] Zhu, B., Zhang, X., Li, H., Liang, J., Wang, R., **Li, H.** and Nishiwaki, S. (2020). "An 89-line code for geometrically nonlinear topology optimization written in FreeFEM." *Structural and Multidisciplinary Optimization*, 63, 1015–1027. 10.1007/s00158-020-02733-x.

[6] Zhang, H., Ding, X., Wang, Q., Ni, W. and **Li, H.** (2020). "Topology optimization of damping composite material with broadband high damping." *Computers & Structures*, 239, 106331. 10.1016/j.compstruc.2020.106331.

[5] **Li, H.**, Ding, X., Meng, F., Jing, D. and Xiong, M. (2019). "Optimal design and thermal modelling for liquid-cooled heat sink based on multi-objective topology optimization: An experimental and numerical study." *International Journal of Heat and Mass Transfer*, 144, 118638. 10.1016/j.ijheatmasstransfer.2019.118638.

[4] **Li, H.**, Ding, X., Jing, D., Xiong, M. and Meng, F. (2019). "Experimental and numerical investigation of liquid-cooled heat sinks designed by topology optimization." *International Journal of Thermal Sciences*, 146, 106065. 10.1016/j.ijthermalsci.2019.106065.

[3] **Li, H.**, Ding, X. and Jing, D. (2019). "Experimental and numerical investigation of fluid cooling channel layout designed by topology optimization." *Journal of Mechanical Engineering*, 055, 198–206. (in Chinese). 10.3901/JME.2019.10.198.

[2] Zhang, H., Ding, X., **Li, H.** and Xiong, M. (2019). "Multi-scale structural topology optimization of free-layer damping structures with damping composite materials." *Composite Structures*, 212, 609–624. 10.1016/j.compstruct.2019.01.059.

[1] Ji, Y., Ding, X., **Li, H.** and Xiong, M. (2018). "Layout design of conductive heat channel by emulating natural branch systems." *Journal of Bionic Engineering*, 15(3), 567–578. 10.1007/s42235-018-0047-3.

Contributions to Books

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Invited Talks

[2] **Li, H.**, "Topology optimization of multiphysics systems." Wuhan University, Mar 8, 2023, University of Shanghai for Science and Technology, Mar 14, 2023. [Link]

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Conference Engagement

Conference Proceedings & Technical Papers

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